FEATURE OVERVIEW

DocuDiff

Validate a customised PDF against its reference — text, formatting, layout, structure, images, links, data formats, accessibility and security — with a dashboard, an interactive inspector, a highlighted PDF and a downloadable report.

16	21	4	33
check groups	issue categories	result views	automated tests passing

01 Validation checks

Run on every comparison. Every check group except OCR is on by default and can be switched off under *Tolerances, checks & rules*.

CHECK	WHAT IT CATCHES	SEVERITY
Missing text	A reference line not found in the customised PDF	Critical / Major
Extra text	A line in the customised PDF that is not in the reference	Major
Changed text	An edited line, with % similarity	Major / Minor
Unfilled placeholders	{{name}}, ${x}, <<x>>, [[x]], [FIRST NAME], %FIELD%, lorem ipsum, XXXX, TODO/TBD	Critical
Spelling	Typos in new text only — reference words and capitalised names are skipped	Minor
Font	Family, bold or italic changed; font substitutions summarised	Major
Font size	Size changed beyond tolerance (default 0.5 pt)	Major / Minor
Text colour	Colour changed beyond tolerance (default 24)	Major / Minor
Fonts present / embedded	Reference fonts missing; fonts not embedded in the file	Major / Minor
Alignment	Left / centre / right changed	Major
Position	Horizontal or vertical shift (default 2 pt)	Major / Minor
Line spacing	Gap between lines changed — reported once, not on every line below	Major / Minor
Margins	Left, top, right, bottom of the text area (default 3 pt)	Major / Minor
Page structure	Page count, missing/extra pages, size, orientation, rotation, image/table/graphics counts	Critical / Major
Table cells	Row/column count and cell-by-cell content	Major

CHECK	WHAT IT CATCHES	SEVERITY
Images	Replaced picture, stretched or distorted, resized, low resolution (below 150 DPI)	Major / Minor
Visual diff	Pixel and structural comparison of rendered pages; unexplained regions flagged	Major / Minor
Hyperlinks	Missing, extra, changed target (old to new URL), malformed URL	Major / Minor
Data formats	Dates, amounts, %, emails, phone, GSTIN, PAN formatted differently; impossible dates; broken emails	Critical / Major / Minor
Bookmarks & form fields	Missing, extra, wrong page, nesting changed; field type or required flag changed	Major / Minor
Accessibility	Tags lost, language not set, images without alt text, low text contrast (WCAG)	Major / Minor
Security & metadata	JavaScript, launch actions, attachments, encryption, PDF/A lost, metadata leaks, title	Critical / Major / Minor
Custom rules	Must contain / must not contain — plain text or /regex/	Critical
OCR (optional)	Reads scanned pages with Tesseract so the text checks work on them	—

02 Scoring and verdict

- **Score** = 35% text + 25% formatting + 15% layout + 25% visual (re-weighted if visual comparison is off).
- **Verdict:** **Failed** on any critical issue or score below 80%; **Needs review** on any major issue or score below 95%; **Passed** otherwise.
- Every issue is rated **critical**, **major** or **minor**.
- Plain-English findings, e.g. *“Font substitution detected: Helvetica → Times-Roman on 1 line(s)”*.

03 Pages

Dashboard

- Stats strip: comparisons, completed, high severity, average similarity
- 14-day trend chart with a Comparisons / Differences toggle
- Severity and verdict donuts, differences by type, quality profile radar, score trend
- Storage usage (size, documents, reports) and recent comparisons with “View all”

New comparison

- Single mode (reference + customised) or batch mode (one reference, many files)
- Drag-and-drop upload; one-click sample PDFs; “What every comparison checks” overview

History

- Every comparison with score bar, verdict and issue count — search, verdict filter, CSV export, open, delete

Batch view

- Live progress for each file, summary counts, open any result

04 Results — four tabs

TAB	WHAT IT SHOWS
Overview	Score ring, verdict, four dimension scores, critical/major count, insights, downloads, Document checks list (pass / warn / fail), page heatmap — click a page to inspect it
Analytics	Five score gauges; donuts for issue type, severity (click a slice to filter issues) and checks; page-by-page similarity; issues per page; top categories; font, size and colour usage
Inspector	Side-by-side PDF viewer with clickable highlight boxes; diff heatmap, before/after slider and blink modes; page navigation; highlights toggle; draw ignore regions and re-run
Issues	Search and filters (severity, category, page, open/accepted); click a row to jump to it; accept / re-open each issue

05 Workflow tools

- **Ignore regions** — exclude areas such as dates or signatures, on one page or all pages
- **Accept as false positive** — remembered for future runs
- **Re-run** — same PDFs with new options
- **Reference library** — save, preview, reuse and delete golden references
- **Presets** — Strict print, Balanced, Loose web, plus your own saved presets
- **Adjustable tolerances** — position, font size, margin, colour, render DPI; on/off switch per check group
- **Keyboard shortcuts** — 1–4 switch tabs, j / k step through issues, ↑ / ↓ and Enter in the Issues table
- **Share link** — read-only view of a comparison
- **Light / dark theme** — remembered per browser

06 Outputs and exports

OUTPUT	CONTENTS
PDF report	Scores, charts, issue table and visual diffs
Highlighted PDF	The customised PDF with boxes coloured red / orange / yellow by severity; dashed blue boxes where reference content is missing
Diff images	One heatmap image per page
JSON	The complete result
CSV	Issues (with open/accepted status) and comparison history

07 Backend and API

AREA	ENDPOINTS / BEHAVIOUR
Jobs	Create (upload or saved reference), status, files, re-run with new options, delete
Batch	Create one reference against many files; check progress
Reference library	List, save, view file, delete
Stats & history	Totals, trends, daily counts, storage usage; every comparison with headline numbers
Safety	Upload size limit and PDF check; protection against path-traversal in file requests
Docs	Interactive API documentation at /docs

08 Running and deployment

- **Local:** SQLite database and files on local disk — no Redis needed
- **Docker Compose:** Postgres, Redis + Celery worker for background jobs, MinIO (S3-style storage), API and web app (nginx); Tesseract OCR included in the image
- **Storage:** local disk, or S3-compatible (AWS S3, MinIO, Cloudflare R2)
- **Database upgrade:** older databases get new columns added automatically on startup
- **Quality:** 33 automated backend tests, all passing; GitHub Actions CI workflow included

09 Sample demo

- Built-in “offer letter” pair — original and customised
- Shows text, font, colour, spacing, link, placeholder, date-format and spelling differences, so every major check is visible in one click